Experiment 7 - The Hall Effect

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1 Background

1.1 Discovery

In 1879, based on work by André-Marie Ampére and James Maxwell, Edwin Hall investigated how magnetic fields interacted with an electrical current in a conductor. He proposed that

$$E' \propto Mv$$

where $E' \equiv V$, $M \equiv |\vec{B}|$ and "v is the velocity of the electricity in the [object]" (E. H. Hall, 1879)[1] (At the time of Hall's paper the electron had only been theorized and wasn't discovered until)

$$V_H = \frac{I_x B_z}{n_e C_e t}$$

where I_x is the current through the conductor, B_z is the magnetic field, n_e is the number of electrons, C_e is the charge of the electron and t is the thickness of the material.

1.2 Hall Effect Sensors

Hall effect sensors Hall effect sensors are often used to measure the position or angle of something, for example a stick in a game controller, the rotation speed or postion of a motor's shaft or measuring current without needing to be inserted directly into a given circuit.

2 References

1. Edwin Hall, 1879, *On a New Action of the Magnet on Electric Currents*, American Journal of Mathematics Vol. 2 No. 3

2. Manny Soltero, 2023, *What Is a Hall-effect Sensor?*, Texas Instruments technical article